

Opening Prayer

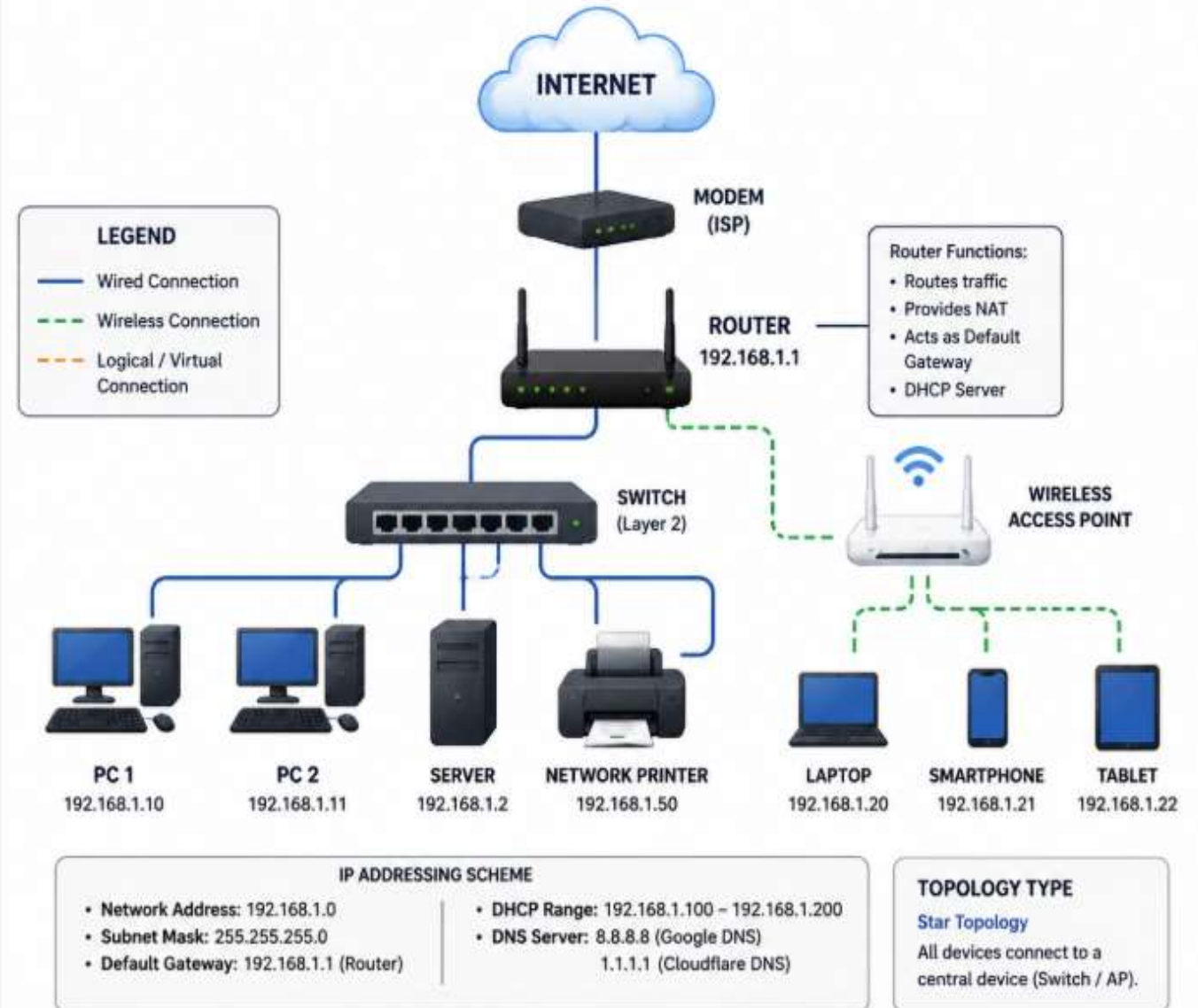


NETWORK CONFIGURATION IN WINDOWS OPERATING SYSTEM

Operating Systems Laboratory
Third Year BSIT / BSCS

NETWORK TOPOLOGY

Example: Office / Small Business Network



LEARNING OUTCOMES

At the end of this lesson, students should be able to:

- ✓ Explain network configuration.
- ✓ Identify basic networking components.
- ✓ Configure IPv4 settings in Windows.
- ✓ Differentiate Static and Dynamic IP addresses.
- ✓ Use Windows networking tools.
- ✓ Perform basic network troubleshooting.

What is Network Configuration?

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Network Configuration is the process of setting up a computer's network settings so it can communicate with other devices and access the Internet.

It includes:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server
- Network Adapter Settings

Why is it Important?

- ✓ Internet Access
- ✓ File Sharing
- ✓ Printer Sharing
- ✓ Remote Access
- ✓ Network Security

PC → Router → Internet

Basic Network Connection



PC
(Client)

Router
(Gateway)



The PC generates data and sends it to the router.



The router receives the data, processes it, and forwards it to the Internet.



The data reaches the Internet and is delivered to the destination.

Data Flow

PC → Router → Internet

Basic Network Components

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IP Address

A unique address assigned to every device.

Example: **192.168.1.100**

Subnet Mask

Determines the network and host portions.

Example: **255.255.255.0**

Basic Network Components

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Default Gateway

Allows communication outside the local network.

Example **192.168.1.1**

DNS Server

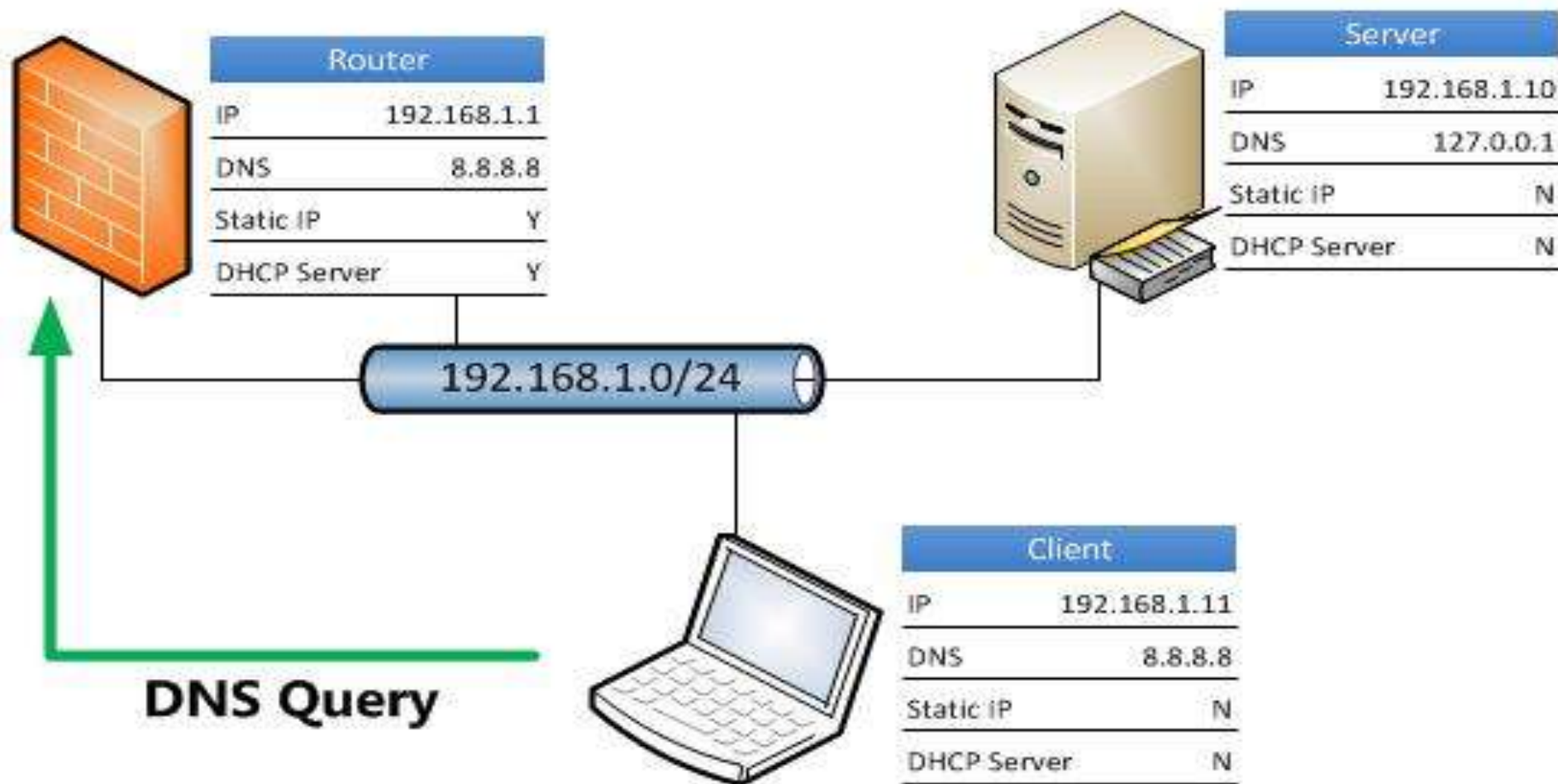
Translates website names into IP addresses.

Google DNS

Example **8.8.8.8**

Cloudflare DNS

Example **1.1.1.1**



Static vs Dynamic IP Address

Static IP

Manual Configuration

Permanent Address

Used by Servers

Less Flexible

Dynamic IP (DHCP)

Automatic Configuration

Temporary Address

Used by Home PCs

Easy to Manage

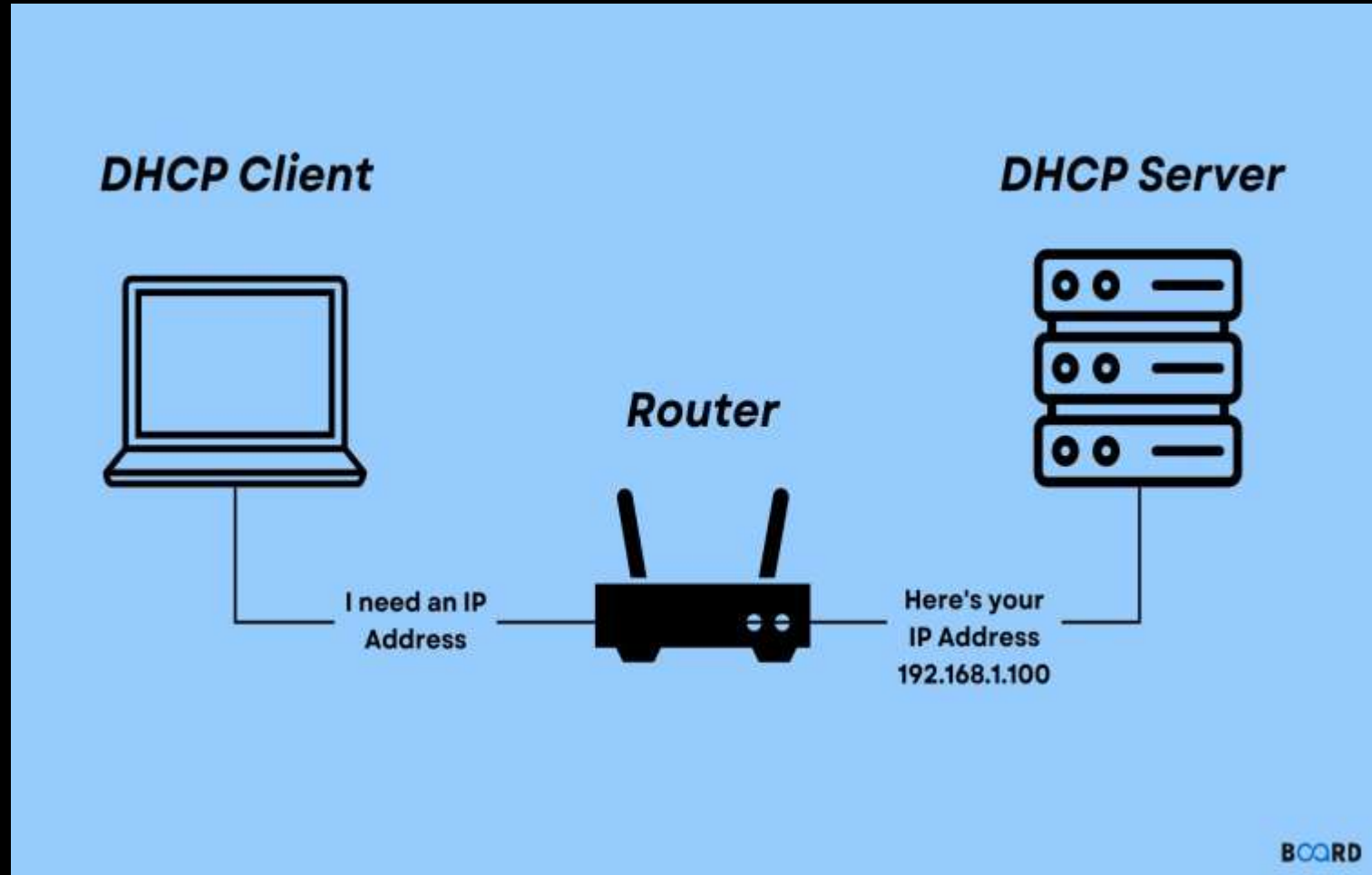
DHCP

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Dynamic Host Configuration Protocol

Automatically assigns:

- IP Address
- Subnet Mask
- Gateway
- DNS

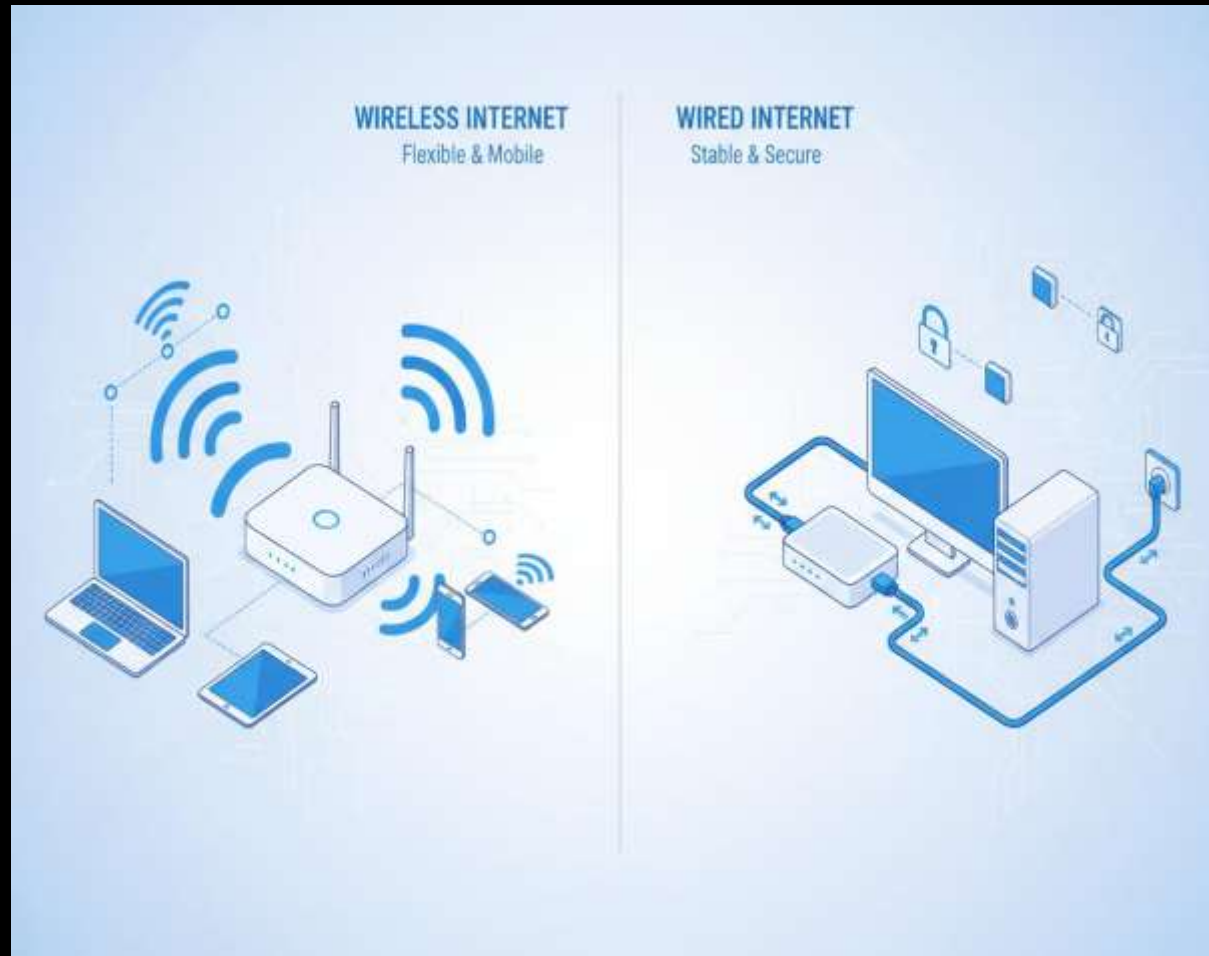


Types of Network Connections

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Wireless Connection

- Wi-Fi
- Mobile
- Convenient



Wired Connection

- Ethernet Cable
- Faster
- Stable

Ethernet (Wired)

🔌 Uses an Ethernet (RJ-45) cable

Faster data transfer

More stable connection

Lower latency

Less interference

Ideal for gaming, servers, and offices

Wi-Fi (Wireless)

📶 Uses radio waves

Convenient mobility

Easy to connect multiple devices

Signal strength depends on distance

Can be affected by walls and other devices

Ideal for laptops, smartphones, and tablets

VIEWING NETWORK CONFIGURATION

1. Open Command

Prompt

Commands:
ipconfig

Displays:

IP Address
Gateway
Subnet Mask

Commands:

ipconfig /all

Displays:

- MAC Address
- DNS
- DHCP
- Host Name

Demonstration

1. Press **Windows + R**.
2. Type **cmd** and press **Enter**.
3. In Command Prompt, type:
4. Point out the IPv4 Address, Subnet Mask, and Default Gateway in the output.
5. Explain how these values help determine whether a computer is properly connected to the network.

Administrator: Command Prompt

Microsoft Windows [Version 10.0.18363.752]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Windows\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix	:	
Link-local IPv6 Address	:	fe80::c507:4b5d:ff0:73f7%15
IPv4 Address	:	10.0.2.5
Subnet Mask	:	255.255.255.0
Default Gateway	:	10.0.2.1

C:\Windows\system32>

NETWORK DIAGNOSTIC COMMANDS

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Administrator: C:\Windows\system32\cmd.exe

```
C:\Users>ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
```

```
Reply from 8.8.8.8: bytes=32 time=14ms TTL=47
```

```
Reply from 8.8.8.8: bytes=32 time=14ms TTL=47
```

```
Reply from 8.8.8.8: bytes=32 time=14ms TTL=47
```

```
Reply from 8.8.8.8: bytes=32 time=16ms TTL=47
```

```
Ping statistics for 8.8.8.8:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 14ms, Maximum = 16ms, Average = 14ms
```

```
C:\Users>
```

Results

- **Reply from** – The destination device responded successfully.
 - **bytes=32** – Size of each packet sent.
 - **time=18ms** – Round-trip time; lower values indicate a faster connection.
-
- **TTL (Time To Live)** – Limits how many network devices (hops) a packet can pass through before being discarded.
-
- **Packets Sent/Received/Lost** – Indicates whether the connection is reliable. A **0% loss** means all packets reached the destination successfully.

Common Ping Results

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Result

Meaning

✓ Reply from...

Connection is successful.

✗ Request timed out

The destination did not respond.

✗ Destination host unreachable

The computer cannot reach the target network or device.

✗ Ping request could not find host

DNS could not resolve the hostname.

DEMONSTRATION:

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1. Open Command Prompt.
2. Type:

```
</> cmd  
ping google.com
```

or

```
</> cmd  
ping 8.8.8.8
```

Observe:

1. Response time (time)
2. Packet loss TTL value.
3. Explain how the ping command is one of the first tools used by IT professionals to verify network connectivity and troubleshoot communication problems.

CONFIGURING IPV4 SETTINGS.

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Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 50 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 50 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

Settings

Edit IP settings

Manual

IPv4

☒ On

IP address: 192.168.1.200

Subnet mask: 255.255.255.0

Gateway: 192.168.1.1

Preferred DNS: 8.8.8.8

Preferred DNS encryption: Unencrypted only

Alternate DNS:

Save Cancel

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

192 . 168 . 21 . 102

Subnet mask:

255 . 255 . 255 . 0

Default gateway:

192 . 168 . 21 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

8 . 8 . 8 . 8

Alternate DNS server:

. . .

☐ Validate settings upon exit

Advanced...

OK

Cancel

Network Connections

Control Panel > Network and Internet > Network Connections

Organize ▾ Disable this network device Diagnose this connection Rename this connection View status of this connection Change settings of this connection

Ethernet
C780
Realtek PCIe GbE Family Controller

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

192 . 168 . 1 . 144

Subnet mask:

255 . 255 . 255 . 0

Default gateway:

192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

8 . 8 . 8 . 8

Alternate DNS server:

8 . 8 . 4 . 4

☐ Validate settings upon exit

Advanced...

OK

Cancel

Edit IP settings

Manual

IPv4

 On

IP address

192.168.0.10

Subnet prefix length

255.255.255.0

Gateway

192.168.0.1

Preferred DNS

192.168.0.1

Alternate DNS

Save

Cancel

Method 1: Using Windows Settings (Windows 11)

1. Open **Settings**.
2. Select **Network & Internet**.
3. Click **Ethernet** or **Wi-Fi**.
4. Select your active network connection.
5. Under **IP Assignment**, click **Edit**.
6. Choose **Manual** and enable **IPv4**.
7. Enter the network information, then click **Save**.

Setting

Example Value

IP Address

192.168.1.100

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

Preferred DNS

8.8.8.8

Alternate DNS

1.1.1.1

Legacy Method (Control Panel)

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1. Open **Control Panel**.
2. Go to **Network and Sharing Center**.
3. Click **Change adapter settings**.
4. Right-click the active network adapter and select **Properties**.
5. Double-click **Internet Protocol Version 4 (TCP/IPv4)**.
6. Select either:
 1. **Obtain an IP address automatically** (DHCP), or
 2. **Use the following IP address** (Static IP).
7. Enter the required IP configuration and click **OK**.

- **Dynamic IP (DHCP):** Automatically assigns network settings from a DHCP²⁵ server. Recommended for most home and office users.
- **Static IP:** Manually configured and commonly used for servers, printers, and network devices that require a fixed address.
- **DNS Server:** Converts domain names (such as www.google.com) into IP addresses so users can access websites.

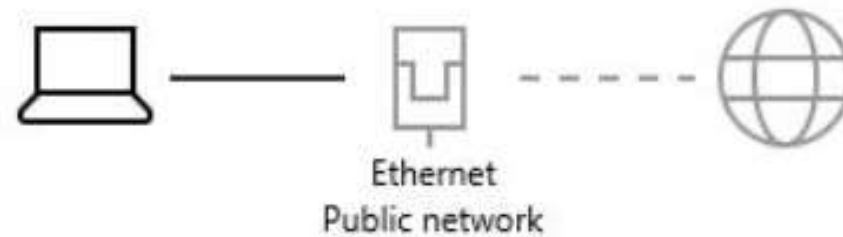
Note: When configuring a static IP address, always ensure it belongs to the correct network and does not duplicate another device's IP address to avoid address conflicts.

Common Network Problems

- ✗ No Internet
- ✗ Wrong IP Address
- ✗ Faulty Cable
- ✗ Disabled Adapter
- ✗ Incorrect DNS
- ✗ Firewall Blocking Connection

Status

Network status



No Internet access

Your device is connected, but you might not be able to access anything on the network. If you have a limited data plan, you can make this network a metered connection or change other properties.

A computer displaying a **network warning icon** indicates that there is a problem with the network connection. The warning icon may appear as a **yellow triangle with an exclamation mark**, a **globe icon**, or a **red X** on the network symbol, depending on the Windows version.

Problem

- ✗ No Internet Connection
- ✗ Limited Connectivity
- ✗ IP Address Conflict
- ✗ Disabled Network Adapter
- ✗ Network Cable Unplugged
- ✗ Incorrect DNS Settings
- ✗ Firewall Blocking Access
- ✗ Outdated Network Driver

Possible Cause

Router or modem is offline, ISP outage
Incorrect IP configuration or DHCP issue
Two devices are using the same IP address
Adapter is disabled in Windows
Ethernet cable is loose or disconnected
Unable to resolve website names
Security settings prevent communication
Driver is missing, corrupted, or incompatible

Class Activity:

If your computer shows a network warning icon and cannot access the Internet, what would be the first three troubleshooting steps you would perform?